

## Determination of Dielectric Constant and Dissipation Factor of a Printed Circuit Board Material Using a Microstrip Ring Resonator Structure

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**Abstract:** A microstrip ring resonator structure is used for determination of dielectric constant and dissipation factor of FR-4 type printed circuit board material. This article consists essential information of the microstrip ring resonator measurement method, characteristic of the microstrip line losses, dielectric constant and dissipation factor calculation techniques. Basing on presented information, dielectric properties of high loss printed circuit board materials can be determined at wide frequency band.

### I. INTRODUCTION

A ring resonator structure is employed in many different microwave and radio frequency circuits. Dielectric properties measurements are an interesting branch of the ring resonator applications. Originally the first ring resonator application, a microstrip ring resonator, was proposed by Troughton [1] for determination of phase velocity and dispersive characteristics of a microstrip line. Since then, ring resonator structures have been used to determine effective permittivity, dielectric constant, dissipation factor and losses in function of the frequency.

Dielectric constant and dissipation factor of printed circuit board can be measured accurately at wide frequency range with a single microstrip ring resonator structure. Benefits of the ring resonator structure on the dielectric properties measurement are:

- 1) The ring resonator structures can be manufactured in normal industrial etching process.
- 2) Minor radiation losses from the resonator structure. There is no need for calculation or measurement of the radiation losses.
- 3) Accurate resonance frequency and harmonics over a wide frequency band.

Dielectric constant of a printed circuit board can be determined by using an one-port or a two-port ring resonator structure. With the two-port ring resonator structure a better signal to noise ratio can be achieved and the dissipation factor of material can be determined, thus the two-port ring resonator is often preferable structure. The two-port ring resonator structure includes feed lines, a closed transmission line loop and coupling gaps. The function of the two-port ring resonator is based on simple equation:

$$2\pi r = n\lambda_g \quad (1)$$

where  $r$  is a mean radius of the ring and  $n\lambda_g$  wavelength or harmonic of wavelength. (For more information, refer [1], [2].)

### II. USE OF THE MICROSTRIP RING RESONATOR STRUCTURE ON DIELECTRIC PROPERTIES MEASUREMENT

Usually in dielectric properties measurements frequency response of the two-port ring resonator structure is measured with a network analyzer. The unwanted effects of connector interfaces of the ring resonator structure, have to be eliminated for example by a thru-reflect-line (TRL) calibration. Basing on the measured frequency response and (1) can be values of effective permittivity calculated in function of frequency. The dielectric constant can be calculated basing on the frequency dependent value of the effective permittivity. Losses of material can be also calculated based on the measured frequency response of the ring resonator. In order to get reliable measurement results, the effects of coupling gaps have to be known or eliminated. In addition, the mean radius of the ring and characteristic impedance of the ring resonator, have to be chosen to be suitable to eliminate curvature effect and higher order modes.

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### A. The Effects of The Coupling Gaps

The effects of the coupling gaps are not taken into account in the equation (1). The coupling gaps have an influence to the resonance frequencies of the ring resonator structure. Normally coupling between the feed lines and the ring resonator are tried to implement with loose coupling. The loose coupling produces negligible effects to the resonance frequencies [2, page 63]. Suitable length for the coupling gaps has to be chosen by using prototypes or modern electromagnetic simulation tools. The effects of the different lengths of coupling gaps are minor to determined values of dielectric constant and dissipation factor, if the coupling is implemented loose.

## III. CALCULATION VALUES OF THE DIELECTRIC CONSTANT AND DISSIPATION FACTOR

The basic idea behind the determination the values of the dielectric constant, is design the microstrip ring resonator to the certain main resonance frequency. The design is based on the estimated value of the dielectric constant. The main resonance frequency and harmonic resonance frequencies will be deviated depending amount of the difference between the estimated and the actual value of the dielectric constant of the material.

An accurate way to calculate values of the dielectric constant at the resonance frequencies can be carried out with an iteration method. The recommend iteration method is based on the Kirschning's and Jansen's [4] accurate model for effective dielectric constant of microstrip. The measured resonance frequencies, the estimated value of dielectric constant, dimensions of the microstrip line and calculated values of the frequency dependent effective permittivity are initial data for the iteration.

### A. Dissipation Factor

Determination of the values of the dissipation factor in function of the frequency, is based on measuring the loaded quality factor  $Q_L$  of the microstrip ring resonator structure at each resonance frequency. The loaded quality factor  $Q_L$  can be obtained from -3dB bandwidth of a resonance peak [2]:

$$Q_L = \frac{f_0}{BW_{-3db}} \quad (2)$$

The unloaded quality factor  $Q_0$ , which includes the total losses the microstrip ring resonator can be calculated from the measured value of the loaded quality factor [2]:

$$Q_0 = \frac{Q_L}{(1 - 10^{-L/20})} \quad (3)$$

For the calculation of the unloaded quality factor is value of the insertion loss  $L$  [dB] needed at each resonance frequency. The total quality factor of the microstrip line includes quality factors of the conductor losses, the dielectric losses and the radiation losses [5]:

$$\frac{1}{Q_0} = \frac{1}{Q_C} + \frac{1}{Q_D} + \frac{1}{Q_R} \quad (4)$$

The radiation losses of the microstrip ring resonator structure are negligible, because there are no open ends in the microstrip ring resonator structure [5]. Thus dielectric losses of the microstrip ring resonator can be calculated basing on the subtract of the total losses of the ring resonator structure and the conductor losses.

$$\alpha_d = \alpha_{tot} - \alpha_c = \frac{8.686\pi}{Q_0 \cdot \lambda_g} - \alpha_c \quad (5)$$

Dissipation factor  $\tan\delta$  can be calculated from the definition of the dielectric losses. Schneider [6] has been presented experimental definition for the dielectric losses:

$$\alpha_d = 27.3 \cdot \frac{\epsilon_r - 1}{\epsilon_r + 1} \cdot \frac{\epsilon_r}{\epsilon_r} \cdot \frac{\tan\delta}{\lambda} \quad [\text{dB/lenght unit}] \quad (6)$$

The conductor losses of the microstrip ring resonator structure can be obtained by calculating, simulating or measuring. Accurate numerical experimental equations for conductor losses have been presented by Pucel et al.[7] and Schneider [8]. The surface roughness of conductors increases the conductor losses. Use of the Morgan experimental expression with conjunction of the equation of Pucel et al. and Schneider to take account the effects of surface roughness, has been recommend by Hammerstad [9]. Conductor losses calculated using either of the equations conjunction with Morgan expression are almost similar.

## IV. EXPERIMENTAL RESEARCH

Experimental research of dielectric properties of FR-4 type material was realized with a microstrip ring resonator structure. In addition, the effects of the calculation equations of the conductor losses to results of the measurement method were investigated. Dimensions of the microstrip ring resonator structure are presented in table 1.

TABLE 1. DIMENSIONS OF THE MICROSTRIP RING RESONATOR STRUCTURE

|                                                |             |
|------------------------------------------------|-------------|
| The main resonance frequency                   | 1,0 GHz     |
| The mean radius of ring resonator              | 26,499 mm   |
| The line width of the ring resonator           | 2,282 mm    |
| The thickness of the substrate material        | 1,195 mm    |
| The length of the feed lines                   | 30,0 mm     |
| The length of the coupling gaps                | 200 $\mu$ m |
| The estimated value of the dielectric constant | 4,3         |
| The thickness of the conductor metal           | 35 $\mu$ m  |

The results of experimental research are presented in fig. 1 and fig 2.

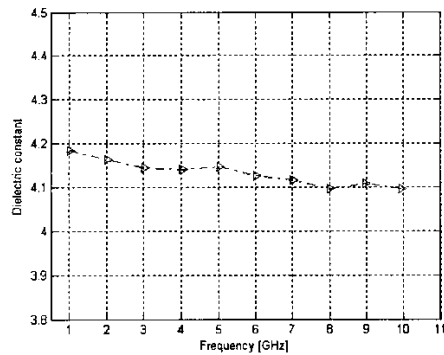


Fig. 1. Values of the dielectric constant in function of the frequency. The overall accuracy of the determination of the dielectric constant is 0.05.

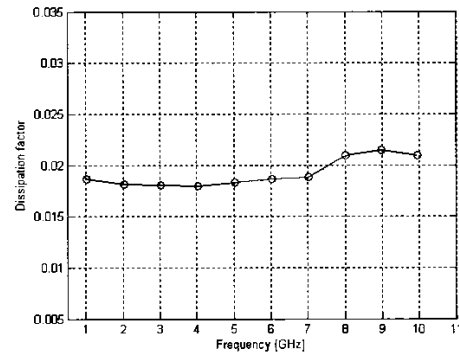


Fig. 2. Values of the dissipation factor in function of frequency. The overall accuracy of the determination of the dissipation factor is 0.005.

The conductor losses of the microstrip ring resonator structure were calculated with equations of Pucell et al. [7] and equations of Schneider [9]. The expression of Morgan was used to take effects of surface roughness of the conductor account. The calculated conductor losses were compared to results gained by ADS/Linecalc program. All those methods are based on Wheeler's incremental inductance rule. In fig. 3 is presented difference between the results of the different conductor losses calculation methods. In fig. 4 is presented effects of the different conductor losses calculation methods to the determined values of dissipation factor.

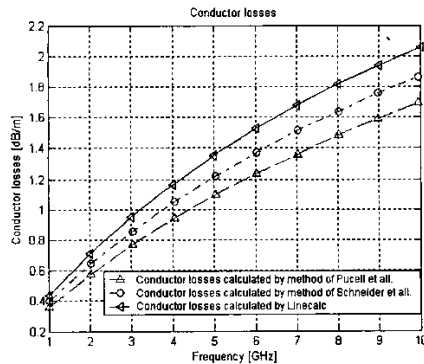


Fig. 3. The conductor losses, calculated by different calculation methods.

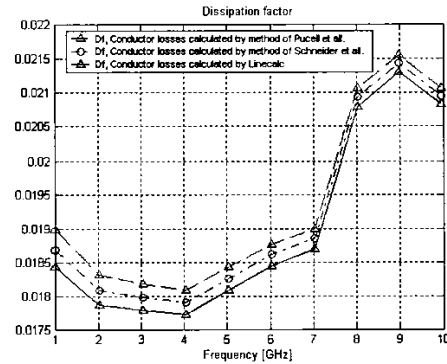


Fig. 4. Effects of the different conductor losses calculation methods to the values of dissipation factor.

The difference between values of the conductor losses calculated by the different equations have only minor effects to calculated values of dissipation factor. Reasons for minor effects of the conductor losses are high dielectric losses of FR-4 type material compared to conductor losses. With low loss microwave material, conductor losses might be dominating losses and determination of the conductor losses have to be done as accurate as possible.

#### V. CONCLUSIONS

The microstrip ring resonator structure can be used to accurate determination of dielectric constant and dissipation factor of printed circuit board materials at wide frequency band. The experimental research with FR-4 type material has shown the method to be suitable for high loss dielectric materials. Important notes are:

- 1) The iteration method based on Kirschning's and Jansen's accurate model for effective dielectric constant of microstrip is an accurate method to determinate dielectric constant of printed circuit board material.
- 2) The difference between the conductor losses calculation methods of Pucell and Schneider has minor effects to the determination of dissipation factor of a high loss printed circuit board material.

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